The Nesting Domain Handoff Problem

Registered Ablation, Parent–Child Context, and Multiscale Gravitational Scenes

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Abstract

Every practical gravitational computation begins by declaring a scene: a domain, source roster, solved geometry, registration convention, approximation regime, observer or causal-access condition, parent context, and uncertainty record. In multiscale systems, a locally resolved child scene is interpreted within a larger parent scene that may be coarser, incomplete, differently registered, or temporally stale. The nesting domain handoff problem is the problem of formalizing that relation without treating either scene as an isolated universe or assigning source-owned parcels to a seamless gravitational field.

This revision replaces the original scalar contribution sums with registered tensor ablation. For a child source or group A, the primary source-conditioned evidence object is the registered change in the electric part of the Weyl tensor,

$$\Delta_A E_{ij} = E_{ij}^{(\mathrm{full})} - \widetilde{E}_{ij}^{(-A)},$$

with Frobenius amplitude under a declared reference inner product as the default direction-agnostic read. Child–parent parity, dominance margins, continuous tensor-allegiance witnesses, parent-context uncertainty, perturbation persistence, and numerical boundary inheritance are assembled into a multichannel handoff record. No universal handoff functional or confidence algebra is claimed. The framework instead separates measured channels from their calibrated aggregation and distinguishes diagnostic modularity from any discontinuity in the physical field.

The Earth–Moon–Sun system remains a minimal example: Earth may organize global acceleration, the Moon may organize local tidal structure, and the Earth–Moon group may retain a distinct registered relation to the solar parent scene. Continuity-preserving refinement and boundary-mismatch measures connect the conceptual handoff problem to future multiscale numerical work. The proposal is not new gravity. It is a disciplined grammar for how solved gravitational scenes nest, inherit context, and preserve relational structure across scale.

Keywords: general relativity; electric Weyl tensor; registered ablation; Frobenius norm; nested scenes; parent context; multiscale gravity; boundary inheritance; relational signature; coarse graining.

1 Introduction

General relativity supplies the local geometric law. Practical computation supplies something else before that law is evaluated: a declaration of what scene is being solved.

A scene is not the universe. It is a bounded and typed claim of relevance. The modeler chooses a source roster, domain, foliation or time slice, approximation regime, coordinates, boundary conditions, field witnesses, registration conventions, and an enclosing context. In multiscale systems, these declarations are nested. A moon is resolved within a planet system; a planet system within a stellar scene; a stellar neighborhood within a galaxy. Each child scene may be locally precise while its parent context is coarser, incomplete, or differently represented.

The nesting domain handoff problem asks:

How should a locally resolved gravitational scene be registered, compared, and inherited within a larger parent scene without pretending that the field itself is partitioned into sovereign pieces?

The revised Gravitational Coherence Surface program provides the machinery required to sharpen this question. Paper I defines registered source–context parity through electric-Weyl ablation. Paper II constructs many-source networks from Frobenius amplitudes of source-conditioned signatures. Paper III studies persistence and relational loss before coarse graining. Paper IV preserves an extraction program and its certification agenda. The Gravitational Field-Relation Operator supplies a direct residual-to-geometry method. The present paper asks how those objects inherit meaning across nested scene declarations.

The proposal is not that the gravitational field is modular as a physical discontinuity. The proposal is that a continuous solved geometry may admit several lawful registered descriptions, and that the transfer of information, uncertainty, and relational structure between those descriptions deserves explicit mathematics.

A context calculus may therefore be useful: not a replacement for GR, but a grammar for parenthood, child refinement, registration, uncertainty, parity, handoff, and boundary inheritance.

2 Scope and claim discipline

This paper does not introduce a new force, field equation, particle, causal horizon, material membrane, or preferred decomposition of spacetime. It does not claim that arbitrary computational boundaries possess invariant physical reality. It does not assert a universal handoff functional, confidence algebra, or identity invariant.

The paper makes a narrower set of claims:

1. Scene declaration can be represented as a formal mathematical object rather than informal bookkeeping.
2. Registered ablation provides a lawful child-conditioned evidence object inside a solved many-source scene.
3. Parent uncertainty, parity, tensor organization, persistence, and numerical inheritance are distinct handoff channels and should not be collapsed into one unexamined scalar.
4. A continuous field can support diagnostic modularity without physical discontinuity.
5. Handoff-aware diagnostics may eventually aid refinement placement, reduced-order modeling, interpolation, or boundary inheritance, but that numerical utility remains a testable research hypothesis.

3 Registered gravitational scenes

Definition 1 (Declared gravitational scene)**.** *A declared gravitational scene is the tuple*

$$\mathfrak{S}(t) = \big(\Omega, \mathcal{R}, \mathcal{G}, \mathcal{I}, \gamma_{\mathrm{ref}}, \mathcal{O}, C_O, \Pi, \Lambda\big), \tag{1}$$

where Ω is the declared domain, $\mathcal{R}$ is the source and intervention roster, $\mathcal{G}$ is the solved geometry or field bundle, $\mathcal{I}$ is the point and frame registration convention, γ_{ref} is the reference spatial inner product, $\mathcal{O}$ is *the witness and relation-operator family, C_O is the causal-access declaration, Π is the parent-context declaration, and Λ is the uncertainty and confidence ledger.*

A scene declaration separates three layers.

Physical construction. The field or geometry is solved under declared equations, approximations, matter assumptions, boundary conditions, and masks.

Registration. Points and frames are identified between full, intervention, child, and parent scenes before tensor subtraction or inheritance is attempted.

Interpretive relation. A declared witness or residual asks a specific question of the registered evidence objects.

A scene is therefore partly physical and partly declared. The curvature is not arbitrary. The question asked of it is.

The observer or causal-access term C_O is operational rather than subjective. In the present slice-based formulation it records available signals, temporal freshness, and admissible reconstruction assumptions. Full light-cone propagation and causal parent updates belong to the dynamical extension.

A scene is a disciplined claim of relevance.

4 Nested scenes and the handoff problem

Let $\mathfrak{S}_A$ be a locally resolved child scene and $\mathfrak{S}_\Pi$ an enclosing parent scene. Their domains, rosters, resolutions, field bundles, registrations, and uncertainties need not match. A child can be internally sharp while its parent is only broadly known. The relation between them has its own confidence and cannot be identified with either scene alone.

Write

$$L_{\text{child}} \neq L_{\text{parent}}, \qquad L_{\text{handoff}} \neq L_{\text{child}}, L_{\text{parent}}.$$

Rather than privilege a multiplicative rule, define a general handoff-confidence functional

$$L_{\text{handoff}} = \mathcal{C}(L_{\text{child}}, L_{\text{parent}}, R_{\text{registration}}, R_{\text{stability}}, R_{\text{boundary}}, R_{\text{coverage}})\,. \tag{2}$$

The arguments of $\mathcal{C}$ should be normalized or otherwise placed on declared comparable scales before aggregation. Multiplicative, minimum-rule, Bayesian, interval, or task-specific forms may be appropriate in different implementations. No universal confidence algebra is claimed here, and L_{handoff} is a reporting confidence rather than a new physical field.

Principle 1 (Nested-scene handoff)**.** *A child scene should not be treated as isolated merely because it is locally resolved, and a parent scene should not be compressed into a smooth background before the child–parent relation has been tested for structure relevant to the declared task.*

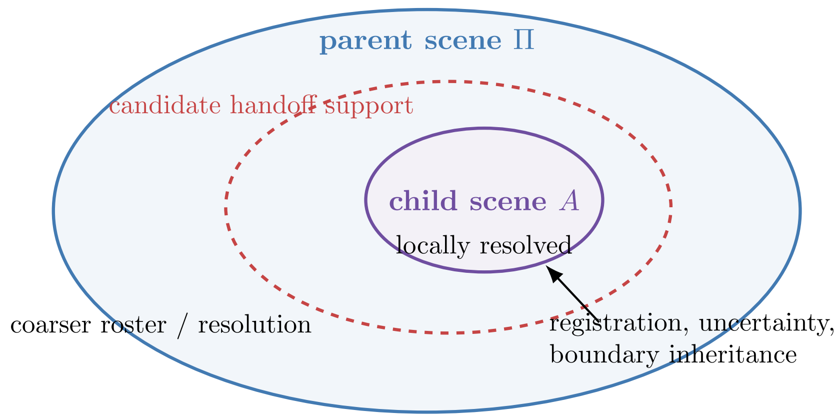

Figure 1: Conceptual nested-scene relation. The handoff support is a diagnostic region defined by registered evidence, confidence, and persistence. It is not a physical membrane.

5 Parent-context uncertainty

A parent context may be necessary for interpreting a child scene while remaining unresolved at the child's fidelity. Parent uncertainty can arise from missing sources, coarse resolution, uncertain boundary conditions, temporal staleness, registration ambiguity, or sensitivity to the chosen witness.

Let $U_\Pi(x)$ be a declared upper bound or uncertainty amplitude for unresolved parent structure, and let $C_\Pi(x)$ be the resolved parent-context amplitude under the chosen witness. Define

$$\Lambda_\Pi(x) = \frac{U_\Pi(x)}{C_\Pi(x) + \epsilon}. \tag{3}$$

In the default tensor lane, C_Π may be a Frobenius amplitude of a registered parent-scene electric-Weyl field or another explicitly declared comparison object. The numerator and denominator must carry the same units, ϵ must be reported in those units, and Λ_Π is dimensionless. It is an uncertainty ratio, not a probability.

The numerator is not an oracle. It is a bound derived from catalog completeness, model discrepancy, unresolved mass limits, temporal uncertainty, boundary-condition uncertainty, or other scene-specific priors. If the unresolved roster were fully known, it would no longer be unresolved.

A schematic parent-confidence rule is

$$L_\Pi(x) \propto \frac{1}{1 + \Lambda_\Pi(x)}, \tag{4}$$

but the proportionality and aggregation belong to the declared confidence model rather than to gravity itself.

The child may be known sharply. The parent may be known broadly. The handoff must carry both facts.

6 Group ablation and child-conditioned signatures

The original paper described child and context through scalar source sums. The revised formulation begins with solved scenes.

Let $E_{ij}^{(\mathrm{full})}$ be the electric part of the Weyl tensor on the full declared slice. Let $E_{ij}^{(-A)}$ be the corresponding tensor in a comparison scene obtained by ablating child source or group A. After point and frame registration, write the comparison tensor as $\widetilde{E}_{ij}^{(-A)}$. Define the group-ablation response

$$\Delta_A E_{ij} = E_{ij}^{(\text{full})} - \widetilde{E}_{ij}^{(-A)}. \tag{5}$$

This is the response of the whole solved scene to the presence of A, including changed nonlinear couplings. It is not a privately owned field parcel of the child group.

Using the declared positive-definite reference metric, define the default Frobenius amplitude

$$S_A(x) = \|\Delta_A E(x)\|_{F,\text{ref}}, \tag{6}$$

and the registered comparison-scene contextual amplitude

$$C_A(x) = \|\widetilde{E}^{(-A)}(x)\|_{F,\text{ref}}. \tag{7}$$

The child–context parity residual is

$$\Psi_{A|\Pi}(x) = S_A(x) - C_A(x), \tag{8}$$

with parity set

$$\Sigma_{A|\Pi} = \{x \in \Omega : \Psi_{A|\Pi}(x) = 0\}. \tag{9}$$

The subscript Π records the declared parent interpretation. In a strict full-versus-ablated construction, C_A is the registered ablated-scene comparator rather than an independently extractable remainder already living in the full geometry. When the ablated comparison scene is not identical to the operational parent model, the paper must report that distinction explicitly rather than treating C_A and C_Π as interchangeable.

The GCS is one level set within a generally global source-conditioned response. It is not the edge of the response.

7 Parity, dominance, and modular scene structure

Nested-scene analysis contains several related but distinct objects.

Table 1: Hierarchy of nested-scene diagnostic objects.

Object	Role
Registered ablation tensor $\Delta_A E$	Primary child-conditioned evidence object
Frobenius amplitude S_A	Default direction-agnostic signature read
Child–context parity set $\Sigma_{A\|\Pi}$	Equality between signature and registered comparison-scene context
Many-source dominance network	Relative ordering and ties among source-conditioned amplitudes
Low-margin region	Neighborhood in which dominance is weak or ambiguous
Fixed-skeleton complex difference	Removed, born, and relabeled relational strata under intervention
Handoff support $B_{A\|\Pi}$	Calibrated multichannel region, not defined by parity alone

Modularity is therefore diagnostic organization, not field decomposition. The same continuous geometry can support lawful questions about child conditioning, parent context, source dominance, and scale transfer without being physically cut into compartments.

Where several source-conditioned amplitudes S_a are available, define a local dominance margin

$$m(x) = \frac{S_{(1)}(x) - S_{(2)}(x)}{S_{(1)}(x) + S_{(2)}(x) + \epsilon}, \tag{10}$$

where $S_{(1)} \geq S_{(2)}$ are the two largest amplitudes at x. Low m identifies a transition neighborhood rather than a zero-thickness wall.

8 Relation to established multiscale concepts

The nesting-domain framework sits beside, not above, established tools.

Hill spheres, spheres of influence, Roche structures, and Lagrange points answer dynamical questions in specified orbital approximations. Multipole expansions compress resolved source structure into moments useful for far-field calculation. Tidal tensors and geodesic deviation describe local relative deformation. Averaging and coarse graining produce indispensable effective descriptions. Numerical mesh refinement transfers data between domains of different resolution.

The present framework asks a prior and complementary question: what registered relational structure should be tracked when moving between differently resolved scene declarations?

Averaging is not inherently destructive. It is successful when it preserves the quantities relevant to the declared task. The handoff problem arises when a coarse summary preserves bulk behavior while erasing child-conditioned distinctions that later prove relevant to tensor organization, trajectory reconstruction, uncertainty, or boundary inheritance.

The framework therefore proposes a reporting and testing layer, not a rival field theory.

9 Tensor-safe handoff witnesses

Parity is not the whole handoff story. A child-conditioned amplitude may reach contextual equality before or after the local tensor organization changes allegiance.

Direct principal-eigenvector residuals are not safe as a default computational witness because eigenvectors can exchange labels or become nonunique near repeated eigenvalues. The revised framework therefore prefers continuous tensor contractions.

Let E_{tot} be a declared registered total or reference tensor. Define normalized child allegiance

$$\mathcal{A}_A(x) = \frac{\left(\Delta_A E : E_{\text{tot}}\right)^2}{\|\Delta_A E\|_F^2 \, \|E_{\text{tot}}\|_F^2 + \epsilon}, \tag{11}$$

and parent allegiance

$$\mathcal{A}_\Pi(x) = \frac{\left(E_\Pi : E_{\text{tot}}\right)^2}{\|E_\Pi\|_F^2 \, \|E_{\text{tot}}\|_F^2 + \epsilon}, \tag{12}$$

where $A : B$ is the Frobenius tensor inner product. A continuous handoff witness is

$$H_{\text{align}}(x) = \mathcal{A}_A(x) - \mathcal{A}_\Pi(x). \tag{13}$$

The exact parent reference tensor is scene dependent and must be declared.

Eigenframe rotation remains valuable as an interpretive channel, but it must be paired with a spectral-gap confidence measure. Near degeneracy, a large frame rotation may reflect basis instability rather than a physically robust orientation transition.

Principle 2 (Tensor continuity)**.** *Computational relation residuals should prefer continuous tensor invariants or contractions. Eigenvectors may be interpreted only where spectral separation and registration confidence are sufficient.*

10 Multichannel handoff features and supports

No single scalar is assumed to define handoff universally. Instead define a local handoff feature vector

$$\mathbf{h}_{A|\Pi}(x) = \big(p_{\text{parity}}, m_{\text{dom}}, a_{\text{tensor}}, u_{\Pi}, r_{\text{stability}}, b_{\text{inheritance}}\big), \tag{14}$$

where the entries record parity proximity, dominance margin, tensor allegiance, parent uncertainty, perturbation persistence, and boundary-inheritance quality. A concrete implementation may use, for example, a bounded parity proximity such as $p_{\text{parity}} = \exp[-|\Psi_{A|\Pi}|/\Psi_0]$ and an uncertainty confidence such as $u_\Pi = (1+\Lambda_\Pi)^{-1}$, but every scale and normalization must be declared. The feature vector is the evidence record; the score below is only one possible decision layer.

A calibrated handoff score may then be written schematically as

$$H_{A|\Pi}(x) = \mathcal{H}\big(\mathbf{h}_{A|\Pi}(x);\theta\big)\,, \tag{15}$$

with parameters or decision rules θ declared by the application. A candidate handoff support is

$$B_{A|\Pi} = \{x : H_{A|\Pi}(x) \geq \tau_H\}, \tag{16}$$

subject to minimum confidence, persistence, and coverage requirements. The parity set $\Sigma_{A|\Pi}$ is an equality locus, whereas $B_{A|\Pi}$ is a thresholded multichannel region. A dominance cell is a third object defined by relative ordering among source-conditioned amplitudes. These objects may intersect, but none should be used as a synonym for another.

Equation (15) is scaffolding, not a field law. The measured channels should be reported separately so that aggregation choices remain inspectable and falsifiable.

11 Relational signature across nested scenes

The original paper used identity language too strongly. The revised object is a nested-scene signature record

$$\mathcal{J}_A^{\Pi} = \big(\Delta_A E, \Sigma_{A|\Pi}, B_{A|\Pi}, \Lambda_\Pi, \mathcal{P}_A^{\Pi}\big)\,, \tag{17}$$

where $\mathcal{P}_A^{\Pi}$ records persistence under changes of resolution, registration, scalarization, parent declaration, and controlled perturbation.

A source or child group has an operational relational signature whenever the declared intervention produces structured, reproducible differences. A stronger claim of relational identity would require invariants of (17) to remain recognizable across context, frame, scale, and time.

Persistent identity is therefore a hypothesis about the stability of the signature record, not a consequence of defining it.

12 Earth–Moon–Sun as a minimal example

The Earth–Moon–Sun system remains a useful conceptual example because it separates several kinds of organization.

At broad scale, Earth may dominate gross acceleration across much of an Earth-centered scene. Near the Moon, Moon-conditioned tidal structure may remain locally readable. At a larger scale, the Earth–Moon group participates in a distinct relation with the solar parent context.

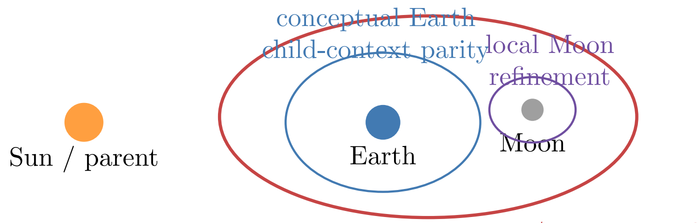

Figure 2: Conceptual Earth–Moon–Sun nesting diagram. The curves illustrate scene roles only; they are not current tensor-ablation outputs, measured boundaries, or physical membranes.

A lawful model should preserve all three statements without flattening them into one scalar ranking:

- Earth can organize global acceleration.
- The Moon can organize local tidal structure.
- The Moon can retain a source-conditioned relational signature within the Earth-parent scene.

The example also clarifies why a local refinement is not merely a cosmetic zoom. A child scene inherits data, uncertainty, and boundary conditions from its parent while resolving structure the parent representation may not carry explicitly.

13 Three dominance channels

Nested scenes become interesting when different notions of dominance diverge.

Acceleration dominance asks which source or group controls gross fall direction or acceleration magnitude.

Tidal-organization dominance asks which source-conditioned tensor structure most strongly organizes local differential deformation.

Relational-signature dominance asks which registered ablation amplitude or signature record is locally strongest under the declared active support and comparison convention.

These channels answer different questions and need not share boundaries. A child source may be globally subdominant in acceleration, locally dominant in tidal organization, and persistently recognizable as a nested relational signature.

The revised language avoids treating relational significance as an already established identity hierarchy.

14 Boundary inheritance and continuity-preserving refinement

A visual grid or numerical mesh is a datum, not a physical substance. Nevertheless, child and parent representations require lawful transfer across their shared boundary.

For a weak-field visualization, a raw release displacement may be written

$$\mathbf{u}_{\mathrm{raw}}(x) = \frac{1}{2}\mathbf{a}(x)\tau^2, \tag{18}$$

with a declared release parameter τ. A continuity-preserving display can be obtained from

$$\mathbf{u}_{\mathrm{cont}} = \arg\min_{\mathbf{u}} \int_{\Omega} \left(\|\mathbf{u} - \mathbf{u}_{\mathrm{raw}}\|^2 + \lambda\|\nabla\mathbf{u}\|^2\right) dV. \tag{19}$$

This is a visualization or interpolation functional, not a gravitational action.

For a child domain Ω_A, let $\mathcal{I}_{\Pi\to A}$ be the declared interpolation or prolongation map from the parent representation. Define a boundary mismatch

$$\mathcal{E}_{\text{handoff}} = \int_{\partial\Omega_A} \left\|\mathbf{u}_A - \mathcal{I}_{\Pi\to A}\mathbf{u}_\Pi\right\|^2 dA. \tag{20}$$

Analogous mismatch measures can be defined for metric data, extrinsic curvature, electric Weyl tensors, constraint residuals, boundary fluxes, or reduced-order state vectors.

This section supplies the paper's clearest bridge to future numerical utility. Handoff-aware geometry may eventually help choose where refinement is placed, what data must be inherited, and which relational distinctions a reduced model must preserve. At present, however, Eq. (20) is a generic interface-mismatch diagnostic. No improvement to stability, convergence, constraint preservation, or computational cost has yet been demonstrated.

15 Claim and evidence register

Table 2: Current claim status.

Item	Status	Scope
Declared nested-scene tuple	Defined	Formal grammar for domain, solve, registration, parent context, and uncertainty
Group-ablation signature $\Delta_A E$	Defined / implemented in revised GCS program	Requires declared solve and registration
FN child–context parity	Defined	Default scalarized handoff anchor, not the whole signature
Parent-context uncertainty ledger	Defined schematically	Calibration remains scene specific
Tensor-safe allegiance witness	Proposed method	Requires benchmark and sensitivity testing
Multichannel handoff feature vector	Defined scaffolding	No universal aggregation rule claimed
Boundary-inheritance mismatch	Standard numerical form adapted here	Utility for numerical-relativity refinement remains prospective and must beat task-matched controls
Persistent nested-scene identity	Hypothesis	Requires invariants across context, frame, scale, and time; not established by the signature record alone
Handoff-aware reduced or multiscale modeling benefit	Research hypothesis	Must outperform task-matched controls

16 Predictions and laboratory agenda

The revised framework generates testable work rather than a finished context algebra.

1. Group-ablation parity sets should persist under controlled changes of resolution, registration convention, norm, and masking when the underlying relation is well resolved.
2. Candidate handoff supports should correlate with continuous tensor-allegiance changes more reliably than with raw principal-eigenvector switches near degeneracy.
3. A child may remain locally signature-dominant while globally acceleration-subdominant.
4. Coarse graining may produce measurable relational loss before large errors appear in bulk mass or far-field potential.

5. Handoff-aware boundary placement or interpolation may improve reconstruction, reduced-order accuracy, or trajectory prediction in selected multiscale tasks.
6. Dynamic slice sequences should reveal whether handoff structures persist, migrate, split, merge, or dissolve.

Immediate laboratory work should begin with static registered slices, then proceed to controlled coarse-graining and boundary-inheritance tests. Any numerical-relativity claim must compare against conventional refinement criteria at matched accuracy, stability, constraint violation, and computational cost.

17 Limits and non-claims

This paper does not claim:

- a unique natural decomposition of every gravitational scene;
- that parity surfaces are physical membranes or dynamical separatrices;
- that FN scalarization exhausts tensor information;
- that handoff supports predict trajectories automatically;
- that eigenframe changes are robust without spectral-gap checks;
- that the feature vector in Eq. (14) has a universal aggregation rule;
- that scene boundaries should replace conventional numerical-refinement criteria;
- that relational identity has been demonstrated;
- or that the framework modifies general relativity.

The framework is a diagnostic and numerical-interface proposal whose value must be earned by stability, comparison, and downstream utility.

18 Discussion: modularity without discontinuity

A modular gravitational scene may sound paradoxical because the geometry is continuous. The paradox dissolves once physical continuity and registered description are separated.

The field does not arrive pre-cut into child and parent territories. A modeler declares scenes because computation, observation, and interpretation have finite scope. Those scenes can be compared lawfully. Their registrations can be audited. Their uncertainties can differ. Their boundaries can inherit data well or poorly. Their ablation signatures can reveal structure that a coarse representation suppresses.

This is modularity without discontinuity.

The local equations may remain correct while inherited bookkeeping of context becomes too coarse for a particular task. Conversely, a handoff-aware model is not automatically superior. If the relational structures fail sensitivity tests or add no predictive value, they should be discarded for that task.

The long-term opportunity is numerical rather than metaphysical. If registered handoff geometry identifies where information is reorganizing, it may guide adaptive interrogation, refinement, reduced representations, or solver continuation. That possibility is ambitious, but it is testable.

19 Conclusion

The nesting domain handoff problem concerns more than where one computational box ends and another begins. It concerns how a locally resolved scene inherits geometry, context, uncertainty, and interpretive structure from an enclosing scene.

The revised framework begins with solved and registered geometries. A child group is ablated, the electric-Weyl response is measured, and the Frobenius amplitude supplies a default parity read. Parent uncertainty, dominance margin, tensor allegiance, persistence, and boundary inheritance are retained as separate channels before any handoff score is constructed.

A continuous gravitational field can therefore admit modular diagnostic structure without being physically partitioned. A child can remain locally meaningful while embedded in a broader parent scene. A parity set can be useful without being a wall. A boundary mismatch can matter numerically without becoming new physics.

The surviving thesis is modest but durable: multiscale gravitational modeling benefits from stating not only what geometry has been solved, but how differently resolved scene declarations are registered, compared, and inherited.

If handoff-aware diagnostics eventually improve refinement placement, reduced-order modeling, or boundary transfer, the context calculus will have moved from descriptive geometry into computational utility. Until then, it remains a disciplined research program with explicit failure conditions.

A Weak-field proxy correspondence

The retired scalar branch remains useful as an analytic correspondence and code check. For a point-source tidal proxy,

$$Q_i(x) = \frac{GM_i}{r_i(x)^3}.$$

For a child group A and a declared parent roster Π, one may define

$$Q_A = \sum_{i \in A} Q_i, \qquad Q_\Pi = \sum_{j \in \Pi} Q_j,$$

and the proxy parity set $Q_A = Q_\Pi$. This lane is frame and decomposition dependent and is not the primary registered-ablation construction.

B Scene passport and confidence ledger

A minimum nested-scene report should include:

- child and parent scene identifiers;
- source and intervention rosters;
- physical regime and solve method;
- point and frame registration maps;
- reference metric and scalarization;
- domain, masks, and active support;
- parent-context uncertainty assumptions;

- root, topology, and coverage confidence;
- boundary interpolation or prolongation map;
- sensitivity to resolution, norm, registration, and roster;
- claim level and known failure conditions.

C Candidate handoff-feature schema

The feature vector in Eq. (14) may be instantiated by:

$$\begin{aligned}
p_{\text{parity}} &= \exp[-|S_A - C_A|/\sigma_p]\,, \\
m_{\text{dom}} &= \frac{S_{(1)} - S_{(2)}}{S_{(1)} + S_{(2)} + \epsilon}, \\
a_{\text{tensor}} &= H_{\text{align}}, \\
u_{\Pi} &= \Lambda_{\Pi}, \\
r_{\text{stability}} &= \text{persistence score under declared perturbations}, \\
b_{\text{inheritance}} &= \exp[-\mathcal{E}_{\text{handoff}}/\sigma_b].
\end{aligned}$$

These expressions illustrate a reportable channel set. They do not define a canonical handoff functional.

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